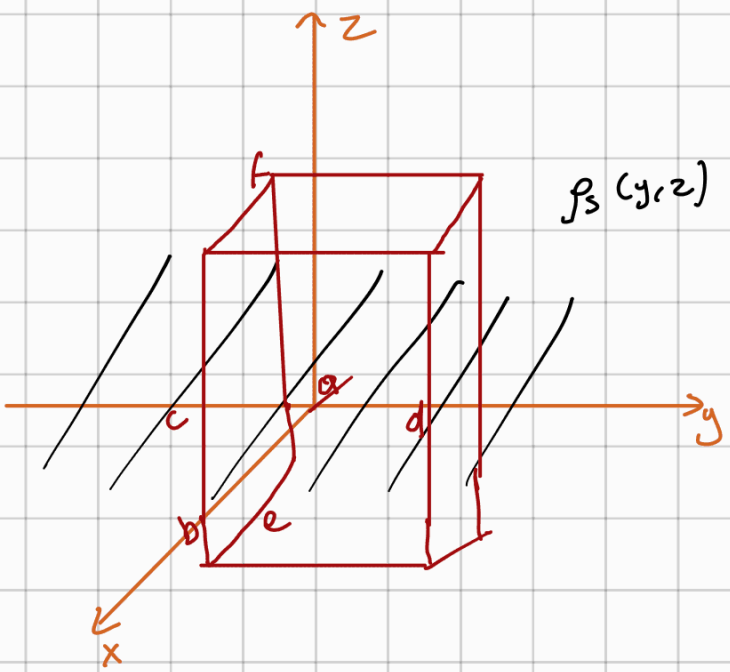
Ders 9

Değişik Türden yük dağılımları ve Dirac Distribüsyonu

hacimsel	yüzeysel	çizgisel
$\rho_v \left[C/m^3 \right]$	$\rho_s \left[C/m^2 \right]$	$\rho_c \left[\frac{C}{m} \right]$



$\rho_s(y,z)$

$x=0$'da $\rho_s(y,z) \left[\frac{C}{m^2} \right]$ yük yoğunluğu olsun

bu yüzeysel yük yoğunluğuna ilişkin hacimsel yük yoğunluğu

$$\underbrace{\rho_v(x,y,z)}_{[C/m^3]} = \underbrace{\rho_s(y,z)}_{[C/m^2]} \underbrace{\delta(x)}_{[1/m]}$$

$\delta(x)$'in özellikleri ♡

1/ $a \le x \le b$, $c \le y \le d$, $e \le z \le f$ prizması içindeki toplam yük ($\rho_v(x,y,z) = \rho_s(y,z)\,\delta(x)$)

$$Q = \int_{x=a}^{b} \int_{y=c}^{d} \int_{z=e}^{f} \rho_v(x,y,z)\, dx\,dy\,dz = \iiint_{x=a\,y=c\,z=e}^{b\,d\,f} \rho_s(y,z)\,\delta(x)\,dx\,dy\,dz$$

$$= \int_{x=a}^{b} \delta(x)\,dx \int_{y=c}^{d} \int_{z=e}^{f} \rho_s(y,z)\,dy\,dz$$

$a, b < 0$, $Q = 0$

$a, b > 0$, $Q = 0$

$a < 0 < b$, $Q = \int_{y=c}^{d} \int_{z=e}^{f} \rho_s(y,z)\,dy\,dz$

$$\int_{x=a}^{0} \delta(x)\,dx = \begin{cases} 0, & ab > 0 \\ 1, & a < x < b \end{cases}$$

2) $\rho_v(x,y,z) = f(x)\,\rho_s(y,z)\,\delta(x)$ olsun

$f(0) = 0$ olursa

$\delta(x)$ nedeniyle $x \neq 0$ $\rho_v = 0$

$f(x)$ nedeniyle $x=0$ $p_v=0$

$f(x)\,\delta(x)=0$ ve $f(0)=0$

$f(x)=g(x)-g(0)$ olsun

$[g(x)-g(0)]\;\delta(x)=0$

$$\boxed{g(x)\,\delta(x)=g(0)\,\delta(x)}$$

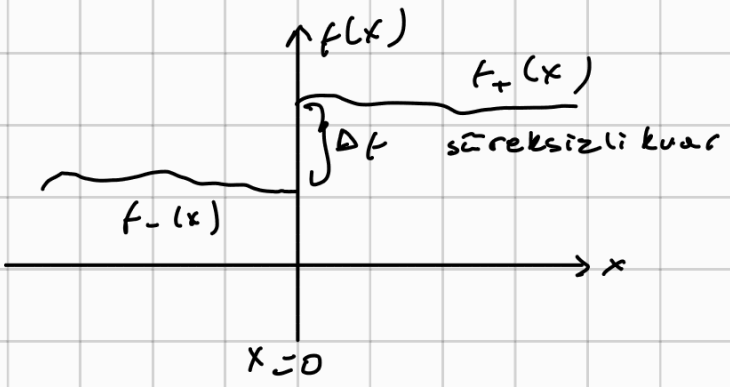
3) $$\int_a^b \delta(x)dx=\begin{cases}0, & ab>0\\ 1, & a<x<b\end{cases}$$

$a\to-\infty$, $b\to x$

$$\int_{-\infty}^{x}\delta(x)dx=\begin{cases}0, & x<0\\ 1, & x>0\end{cases}=u(x)$$ birim basamak fonksiyon

$$\boxed{\delta(x)=\frac{\partial u(x)}{\partial x}}$$

4)



$$f(x)=\begin{cases}f_-(x), & x<0\\ f_+(x), & x>0\end{cases}$$

$x=0$'da süreksizlik kuar

$$f(x)=f_+(x)u(x)+f_-(x)u(-x)$$

$$f'(x)=f_+'(x)u(x)+\underbrace{f_+(x)\delta(x)}_{f_+(0)\delta(x)}+\cancel{f'(x)u(-x)}+f_-'(x)u(-x)$$
$$-\underbrace{f_-(x)\delta(x)}_{f_-(0)\delta(x)}$$

$$f'(x) = \underbrace{\left[f_-'(x)u(-x) + f_+'(x)u(x) \right]}_{\{f'(x)\}} + \underbrace{\left[f_+(0) - f_-(0) \right]}_{[|f|]} \delta(x)$$

$$f'(x) = \{f'(x)\} + [|f|]\,\delta(x)$$

$\{f'(x)\}$: Regüler kısım

$[|f|]\,\delta(x)$: Tekil (singüler kısım)

$\text{grad}\,V = \frac{\partial V}{\partial x}\vec{e}_x + \frac{\partial V}{\partial y}\vec{e}_y + \frac{\partial V}{\partial z}\vec{e}_z$, $x = 0$'da bir süreksizlik olsun

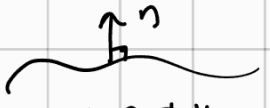
$$\frac{\partial V}{\partial x} = \left\{\frac{\partial V}{\partial x}\right\} + [|V|]\,\delta(x) + [|V\vec{e}_x|]\,\delta(x)$$

$$\frac{\partial V}{\partial y} = \left\{\frac{\partial V}{\partial y}\right\}$$

$$\frac{\partial V}{\partial z} = \left\{\frac{\partial V}{\partial z}\right\}$$



V'nin süreksizliği herhangi bir s yüzeyi üzerinde olsaydı,



$$\text{grad}\,V = \{\text{grad}\,V\} + [|V\vec{n}|]\,\delta(s)$$

5) $x = 0$'da süreksizliği olan $\vec{E}(x,y,z)$ için

$$\text{div}\,\vec{E} = \frac{\partial E_x}{\partial x} + \frac{\partial E_y}{\partial y} + \frac{\partial E_z}{\partial z}$$

$$\frac{\partial E_x}{\partial x} = \left\{\frac{\partial E_x}{\partial x}\right\} + [|E_x|]\,\delta(x)$$

$$\frac{\partial E_y}{\partial y} = \left\{\frac{\partial E_y}{\partial y}\right\}$$

$$+\quad \frac{\partial E_z}{\partial z} = \left\{ \frac{\partial E_z}{\partial z} \right\}$$

$$\text{Div}\,\vec{E} = \{\text{div}\,\vec{E}\} + [|E_x|]\delta(x)$$
$$= \{\text{div}\,\vec{E}\} + [|\vec{E}\vec{e}_x|]\,\delta(x)$$

Genel durumda bir S yüzeyinde süreksizlik varsa

$$\text{div}\,\vec{E} = \{\text{div}\,\vec{E}\} + [|\hat{n}\,\vec{E}|]\,\delta(s)$$
$$\text{rot}\,\vec{E} = \{\text{rot}\,\vec{E}\} + [|\hat{n}\times\vec{E}|]\,\delta(s)$$

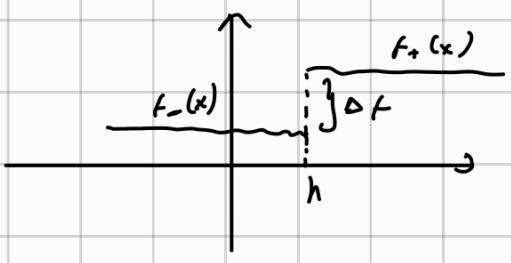
6) Süreksizlik $x=0$ yerine $x=h$'da olursa

$$\delta(x) \to \delta(x-h)$$

$$\int_a^b \delta(x-h)\,dx = \begin{cases} 1, & a<h<b \\ 0, & \text{diğer} \end{cases}$$

$$f(x)\,\delta(x-h) = f(h)\,\delta(x-h)$$

$$\frac{\partial u(x-h)}{\partial x} = \delta(x-h)$$



$$f'(x) = \{f'(x)\} + [|f(h+0) - f(h-0)|]\,\delta(x-h)$$

(Δf)

7) Bir C eğrisi S_1 ve S_2 yüzeylerin ara kesiti olsun. C üzerinde bulunan ρ_c [C/m] için hacimsel yük yoğunluğu

$$\rho_v(x,y,z) = \rho_c\,\delta(S_1)\,\delta(S_2)$$

$S_1 : x = a$
$S_2 : y = b$

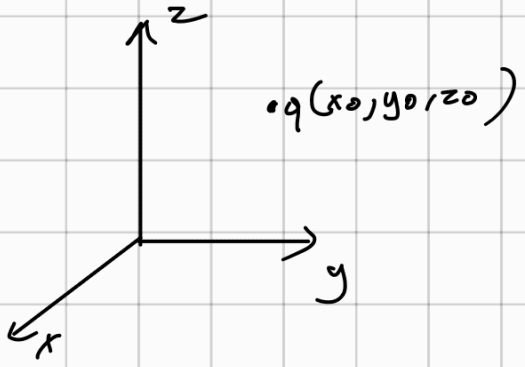
$$\underbrace{\delta(x-a)}_{S_1}\,\underbrace{\delta(y-b)}_{S_2}$$

Örneğin z ekseninde yerleştirilmiş çizgisel yük için

$$\rho_v(x,y,z) = \rho_c\, \delta(x)\,\delta(y)$$

δ) (x_0, y_0, z_0) noktasında bulunan q noktasal yükü için hacimsel yük yoğunluğu

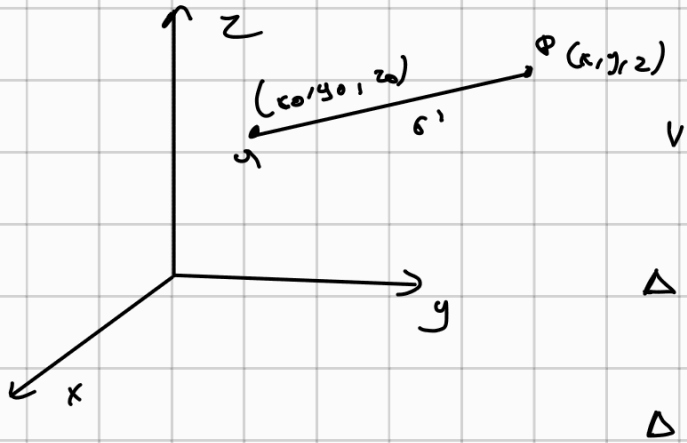
$$\rho_v = q\,\delta(x-x_0)\,\delta(y-y_0)\,\delta(z-z_0)$$



Sıkı Bir

$\Delta\left(\frac{1}{r}\right) = -4\pi\,\delta(x-x_0)\,\delta(y-y_0)\,\delta(z-z_0)$ olduğunu gösteriniz

$$r' = \sqrt{(x-x_0)^2+(y-y_0)^2+(z-z_0)^2}$$



$$V = \frac{1}{4\pi\varepsilon_0}\,\frac{q}{r'}$$

$\Delta V = -\frac{\rho_v}{\varepsilon_0}$ Poisson hacim içinde yük varsa

$\Delta V = 0$ Laplace denklemi hacim içinde yük yoksa

$$\rho_v = q\,\delta(x-x_0)\,\delta(y-y_0)\,\delta(z-z_0)$$

$$\Delta\left(\frac{1}{4\pi\varepsilon_0}\,\frac{q}{r'}\right) = \frac{-q}{\varepsilon_0}\,\delta(x-x_0)\,\delta(y-y_0)\,\delta(z-z_0)$$

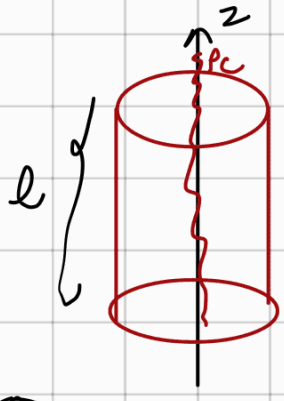
$$\frac{1}{4\pi}\,\Delta\left(\frac{1}{r'}\right) = -\delta(x-x_0)\,\delta(y-y_0)\,\delta(z-z_0)$$

$$\Delta\left(\frac{1}{r_1}\right) = -4\pi\,\delta(x-x_0)\,\delta(y-y_0)\,\delta(z-z_0)$$

silindirik koordinatlardaki uzunluk

$\rho^2 = (x-x_0)^2 + (y-y_0)^2$ olmak üzere

$$\Delta\left(\ln\left(\frac{1}{\rho}\right)\right) = -2\pi\,\delta(x-x_0)\,\delta(y-y_0)$$



$Q = \ell \cdot \rho_c$ (Uzunluğu boyunca)

$$\int_S \varepsilon_0 \vec{E}\cdot d\vec{s} = \ell\rho_c \qquad \vec{E} = E_\rho(\rho)\,\vec{e}_\rho$$

$$d\vec{s} = \rho\, d\phi\, dz\, \vec{e}_\rho$$

$$\int_{z=0}^{\ell}\int_{\phi=0}^{2\pi} \varepsilon_0 E_\rho(\rho)\,\rho\, d\phi\, dz = \ell\rho_c$$

$$2\pi\ell\varepsilon_0\rho E_\rho(\rho) = \ell\rho_c$$

$$E_\rho(\rho) = \frac{\rho_c}{2\pi\varepsilon_0\rho} \Rightarrow \vec{E} = \frac{\rho_c}{2\pi\varepsilon_0\rho}\,\vec{e}_\rho$$

$$\vec{E} = -\text{grad}\,V \Rightarrow \frac{\rho_c}{2\pi\varepsilon_0\rho} = -\frac{\partial V}{\partial\rho}$$

$$\Rightarrow V = \frac{\rho_c}{2\pi\varepsilon_0}\ln\left(\frac{1}{\rho}\right) + V_0$$

$$\Delta V = -\frac{\rho_v}{\varepsilon_0} \qquad \text{ve} \quad \rho_v = \rho_c\,\delta(x-x_0)\,\delta(y-y_0)$$

$$\Delta\left(\frac{\rho_c}{2\pi\varepsilon_0}\ln\left(\frac{1}{\rho}\right)\right) + \Delta V_0 = -\frac{\rho_c}{\varepsilon_0}\,\delta(x-x_0)\,\delta(y-y_0)$$

$$\frac{\rho_c}{2\pi\varepsilon_0}\,\Delta\left(\ln\left(\frac{1}{\rho}\right)\right) = -\frac{\rho_c}{\varepsilon_0}$$

$$\Delta\left(\ln\left(\frac{1}{\rho}\right)\right) = -2\pi\,\delta(x-x_0)\,\delta(y-y_0)$$